TRANSMISSION LINES & WAVEGUIDES – ECA1501

EXPERIMENT – 2

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Date	12/10/2026	Experiment No.	2

Analysis of SWR (Standing wave ratio) Transmission lines by varying a) Load Impedance b) Characteristic Impedance

AIM

To analyze the SWR of a transmission line for a fixed characteristic impedance (Zo) of 50 ohms, as the load impedance (ZL) is varied across the following values: 25 ohms, 50 ohms, 75 ohms, and 100 ohms. To verify how the SWR changes with each value of ZL and when ZL =Zo.

SOFTWAREREQUIRED:

SCILABVersion6.1.1

PROCEDURE:

- Open SCILAB software
- Open file-new
- Put the input values
- Type the program
- Execute the program
- End the program

PROGRAM:

```
clc;clear;close;
Zo=200;ZR=650-(%i*475);
K=(ZR-Zo)/(ZR+Zo);
ampK=sqrt((real(K)^2)+(imag(K)^2));S=(1+ampK)/(1-ampK);
printf("(c)Standing wave ratio = %f",round(S*1000)/1000);
```

TEST CASE 1

Analyze the SWR of a transmission line for a fixed characteristic impedance (Zo) of 50 ohms, as the load impedance (ZL) is varied across 650 ohms to 850 ohms. Verify how the SWR changes with each value ofZL. When ZL=Z0, the SWR should equal 1 (indicating no reflection).

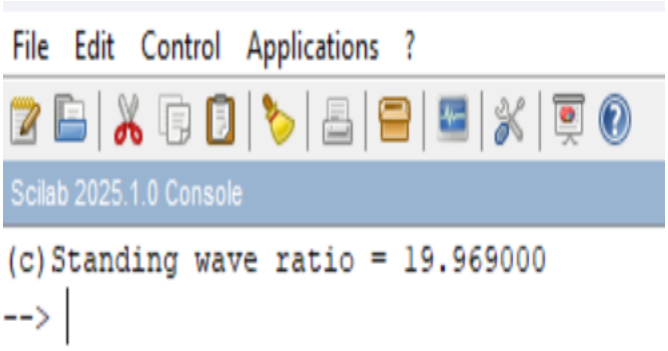
OUTPUT TABLE 1

Zo=50 ohms Variable: ZR

ZR (Ohms)	Simulation Results (SWR)	Theoretical Results (SWR)
50	1.000	1.000
100	2.000	2.000
150	3.000	3.000
200	4.000	4.000
250	5.000	5.000
300	6.000	6.000

ZR=50/40ohmsVariable: Zo

Z0 (Ohms)	Simulation Results (SWR)	Theoretical Results (SWR)
50	2.145	2.145
100	2.921	2.921
150	4.108	4.108
200	5.361	5.361
250	6.637	6.637
300	7.924	7.924



RESULT:

The SWR of a transmission line for a fixed characteristic impedance (Zo) of 50 ohms, is analyzed for various values of load impedance ZL. SWR is also examined for the condition ZL =Zo.

TEST CASE 2

A. Analyze the SWR for a fixed load impedance (ZL) of 50 ohms while varying the

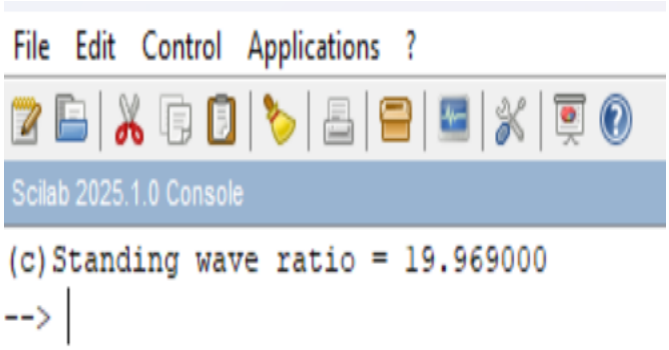
B. characteristic impedance (Z0) of the transmission line to 250 ohms to 400 ohms. Verify how

C. the SWR changes with each value Z0 including Z0=ZL. Variable Resistance (R): L = 3.5 mH/km, C = 8 µF/km, G = 4 µmho/km, f = 2.5 kHz

OUTPUT:

ZR=50/_40o ohms Variable: Zo

Z0 (Ohms)	Simulation Results (SWR)	Theoretical Results (SWR)
50	2.145	2.145
100	2.921	2.921
150	4.108	4.108
200	5.361	5.361
250	6.637	6.637
300	7.924	7.924



RESULT:

SWR for a fixed load impedance (ZL) of 50 ohms is analyzed while varying the characteristic impedance (Z0) of the transmission line to 250 ohms to 400 ohms. Impact of the condition Z0=ZL on SWR is also examined for various values of Z0 .